MATHEMATICS DEPARTMENT
FAITH · RIGHTEOUSNESS · WISDOM

cognia SCHOOL OF DISTINCTION 2024

مدارس دار الفكر
DAR ALFIKR SCHOOLS

GRADE 11 · ALGEBRA II · TOPIC 6

Lesson 6-6

Exponential and Logarithmic Equations and Inequalities

$$2^x = 8 \iff x = \log_2 8$$

Journey of 6 stations, 60 minutes

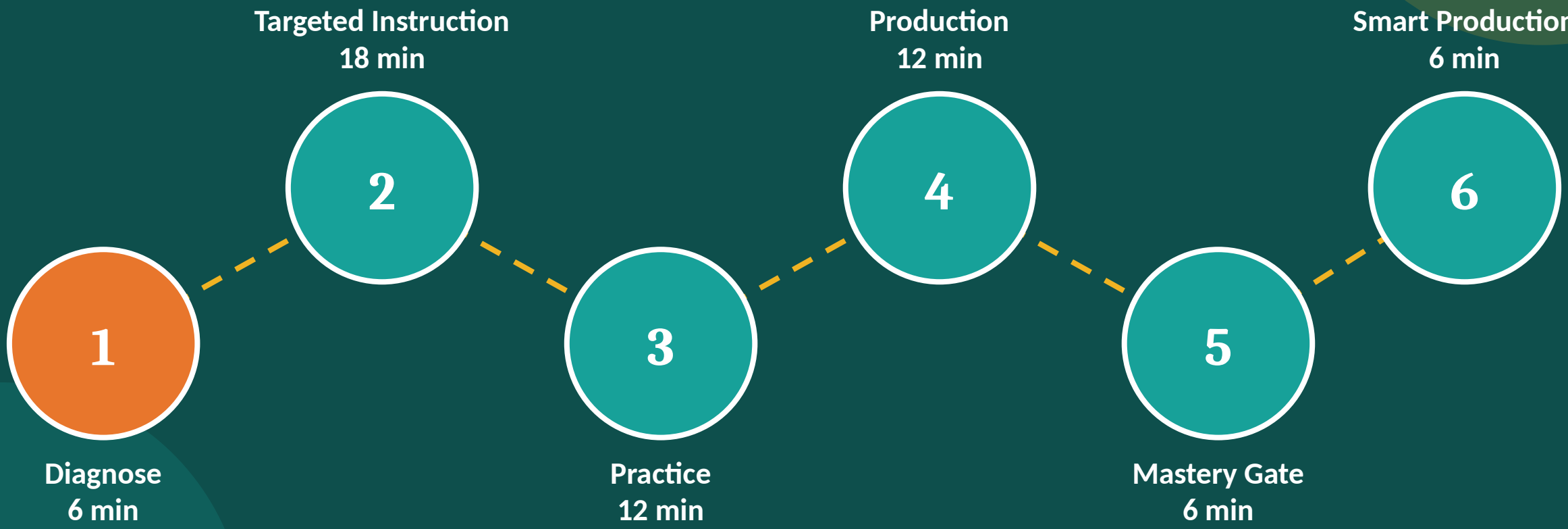
Exponential and Logarithmic Functions

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Today's journey

Six stations. Each one zooms in as we arrive.



What we are learning

OBJECTIVES

1. Solve exponential equations and inequalities by applying the concept of a common base.
2. Solve logarithmic equations and inequalities by using the rules of logarithms.
3. Use logarithms to solve exponential equations and models.

ESSENTIAL QUESTION

How can we solve exponential and logarithmic equations and inequalities?

MATHEMATICAL PRACTICES MP.2 · MP.3

VOCABULARY

- exponential equation
- exponential inequality
- logarithmic equation
- logarithmic inequality

STANDARDS

HSA.SSE.A.2 Use the structure of an expression to identify ways to rewrite it.

HSA.CED.A.1 Create equations and inequalities in one variable and use them to solve problems. Include equations arising from linear and quadratic functions, and simple rational and exponential functions.

HSA.REI.A.1 Explain each step in solving a simple equation as following from the equality of numbers asserted at the previous step, starting from the assumption that the original equation has a solution. Construct a viable argument to justify a solution method.

HSF.LE.A.4 For exponential models, express as a logarithm the solution to $ab^{ct} = d$ where a, c, and d are numbers and the base b is 2, 10, or e; evaluate the logarithm using technology.

STATION 1 · DIAGNOSE · 6 MIN

Start here: three quick checks

1

Solve $5^x = 125$

2

Write 16^x as a power of 2

3

Evaluate $\log_3 81$ and $\log_2 \frac{1}{8}$

Done when

All three answers are written without a calculator, and you can say why 2 to the power 4x is the same as 16 to the power x.

STATION 2 · TARGETED INSTRUCTION · 18 MIN

Exponential equations: match the bases

$$b^m = b^n \iff m = n$$ write both sides with one common base

EXAMPLE A

Solve $2^{x+1} = 32$

$$2^{x+1} = 2^5$$

$$x + 1 = 5$$

$$x = 4$$

EXAMPLE B

Solve $9^x = 27^{x-1}$

$$3^{2x} = 3^{3x-3}$$

$$2x = 3x - 3$$

$$x = 3$$

EXAMPLE C

Solve $\left(\frac{1}{4}\right)^x = 8^{x-2}$

$$2^{-2x} = 2^{3x-6}$$

$$-2x = 3x - 6$$

$$x = \frac{6}{5}$$

STATION 2 · TARGETED INSTRUCTION · 18 MIN

Exponential inequalities: check the base

$$b > 1: \quad b^m < b^n \iff m < n \qquad 0 < b < 1: \quad b^m < b^n \iff m > n \ \text{(flip)}$$

EXAMPLE A

Solve $3^{2x-1} > 27$

$3^{2x-1} > 3^3$

$2x - 1 > 3$

$x > 2$

EXAMPLE B

Solve $\left(\frac{1}{2}\right)^x < \frac{1}{16}$

$\left(\frac{1}{2}\right)^x < \left(\frac{1}{2}\right)^4$

base $\frac{1}{2} < 1$, so flip

$x > 4$

EXAMPLE C

Solve $4^x \leq 8^{x-1}$

$2^{2x} \leq 2^{3x-3}$

$2x \leq 3x - 3$

$x \geq 3$

STATION 2 · TARGETED INSTRUCTION · 18 MIN

Logarithmic equations: rules, then check

$$\log_b M + \log_b N = \log_b(MN) \qquad \log_b M = \log_b N \Rightarrow M = N \qquad \log_b x = c \iff x = b^c$$

EXAMPLE A

Solve
$\log_2(x+3) + \log_2(x-3) = 4$

$\log_2\left(x^2 - 9\right) = 4, \quad x^2 = 25$

$x = \pm 5,$ but need $x > 3$

$x = 5 \quad (-5$ is extraneous$)$

EXAMPLE B

Solve $\log_5(3x+1) = 2$

$3x + 1 = 5^2 = 25$

$x = 8$

check: $3(8) + 1 = 25 > 0$ ✓

EXAMPLE C

Solve $\log_2(x-1) < 3$

$x - 1 > 0$ and $x - 1 < 2^3$

$1 < x < 9$

STATION 2 · TARGETED INSTRUCTION · 18 MIN

Use logarithms when the bases will not match

$$b^x = c \iff x = \log_b c = \frac{\ln c}{\ln b} \qquad e^{kt} = c \iff t = \frac{\ln c}{k}$$

EXAMPLE A

Solve $2^x = 20$

$$x = \log_2 20$$

$$x = \frac{\ln 20}{\ln 2} \approx 4.32$$

EXAMPLE B

Solve

$$1000e^{0.05t} = 3000$$

$$e^{0.05t} = 3$$

$$0.05t = \ln 3$$

$$t = \frac{\ln 3}{0.05} \approx 22.0$$

EXAMPLE C

Solve $5^x > 40$

$$x > \log_5 40$$

$$x > \frac{\ln 40}{\ln 5} \approx 2.29$$

STATION 3 · PRACTICE · 12 MIN

Give it a go: Practice, Apply, Investigate

PRACTICE

Secure the method

Solve $3^{x-2} = 81$

Solve $\log_4(x+2) = 3$

Done when: both sides share one base before the exponents are equated, and each logarithm answer is checked in the original equation.

APPLY

Use it in context

Solve $\left(\frac{1}{3}\right)^x > 9$

Solve
$\log_3 x + \log_3(x-2) = 1$

Done when: the direction of the inequality is justified by the size of the base, and each logarithm solution is tested against the domain.

INVESTIGATE

Find out why

Compare the solutions of
$\log_2(x^2) = 2$ and $2\log_2 x = 2$

Solve $\log_{1/2} x > 2$ and say why the sign flips

Done when: your explanation uses the domain of the logarithm, and you test your inequality with one value inside and one outside your answer.

STATION 4 · PRODUCTION · 12 MIN

Production: Red Sea resort visitor targets

THE MODEL

Vision 2030 includes new Red Sea tourism resorts. Using model values only (not forecasts), two resorts are modelled by the visitors-per-year functions below, with time measured in years after opening. The tourism board wants to know when each resort reaches its targets.

$$A(t) = 40\,000\,(1.25)^t$$

$$B(t) = 20\,000 \cdot 2^{t/3}$$

1. Find when Resort A reaches 125 000 visitors: write it as a logarithm first, then give the time to 1 decimal place.
2. Find when Resort A first exceeds 200 000 visitors, and say how many whole years are needed.
3. Use a common base to find when Resort B reaches 160 000; then find when Resort A does, and say which resort is first.

DONE WHEN

- Task 1 is written as a logarithm first and evaluated: about 5.1 years.
- Task 2 gives a time above about 7.2 years, so 8 whole years, with the inequality direction kept.
- Task 3 matches bases for Resort B: the exponent must equal 3, giving 9 years.
- The comparison says Resort A is first: about 6.2 years against 9 years.

Hint for solving: if the bases cannot be matched, take the natural logarithm of both sides; if they can, match them.

STATION 5 · MASTERY GATE · 6 MIN

Mastery Gate: three questions to pass

1

Solve $4^{x+1} = 8^x$

2

Solve
$\log_3 x + \log_3(x+6) = 3$

3

Solve $\left(\frac{1}{2}\right)^x > 8$

Pass mark

3 of 3 correct unlocks Station 6. Fewer than that: return to the slide for the question you missed, redo it, then re-take the gate.

STATION 6 · SMART PRODUCTION · 6 MIN

Smart Production: make one, then check it

CREATE

1. Write one logarithmic equation whose algebra gives two candidate solutions, where exactly one is valid.
2. Write the full solution on the back, with the domain check and the rejected value.
3. Swap with a partner and solve theirs.

Done when: your equation has exactly one valid solution, and your solution shows the domain check and one test by substitution.

TIME TO CHECK

Exit ticket: write your answer on a sticky note and place it on the door.

Solve

$$\log_2 x + \log_2(x-2) = 3$$

Solve $7^x = 30$ to 3 s.f.

Next lesson: 6-7 Geometric Sequences and Series.

Exam connections

SAT

If $9^{x-1} = 27^{x/2}$, what is x?

STEPS

1 $3^{2x-2} = 3^{3x/2}$

2 $2x - 2 = \dfrac{3x}{2}$

3 $x = 4$

GAT

Which quantity is greater?

A: a if $2^a = 8$ B: b if $4^b = 8$

STEPS

1 $a = 3 \quad (2^3 = 8)$

2 $2^{2b} = 2^3 \Rightarrow b = \dfrac{3}{2}$

3 $3 > \dfrac{3}{2} \Rightarrow$ A is greater

SAAT

Solve $\log_2 x + \log_2(x-1) = 1$

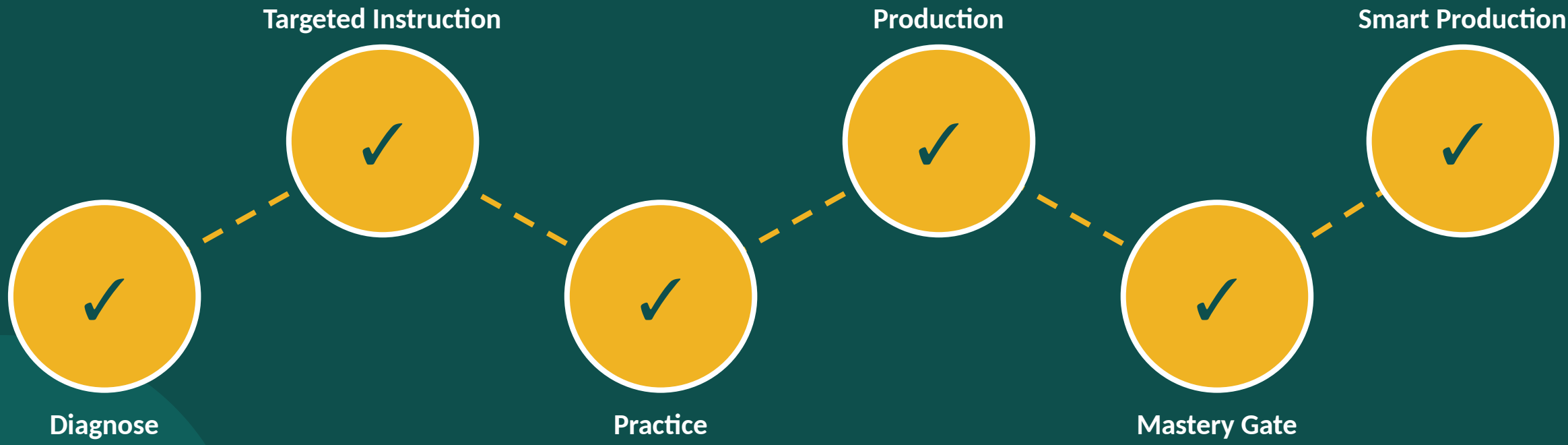
STEPS

1 $x(x-1) = 2 \Rightarrow x = 2$ or -1

2 domain: $x > 1$

3 $x = 2$

Journey complete



To solve an exponential equation, match the bases or take a logarithm; to solve a logarithmic equation, use the rules and then check every answer in the domain; flip an inequality only when the base is below 1.